

Update IOS(ManagementServer)

College of Science, Engineering, and Technology, Grand Canyon University

ITT-216: Enterprise Route & Switch

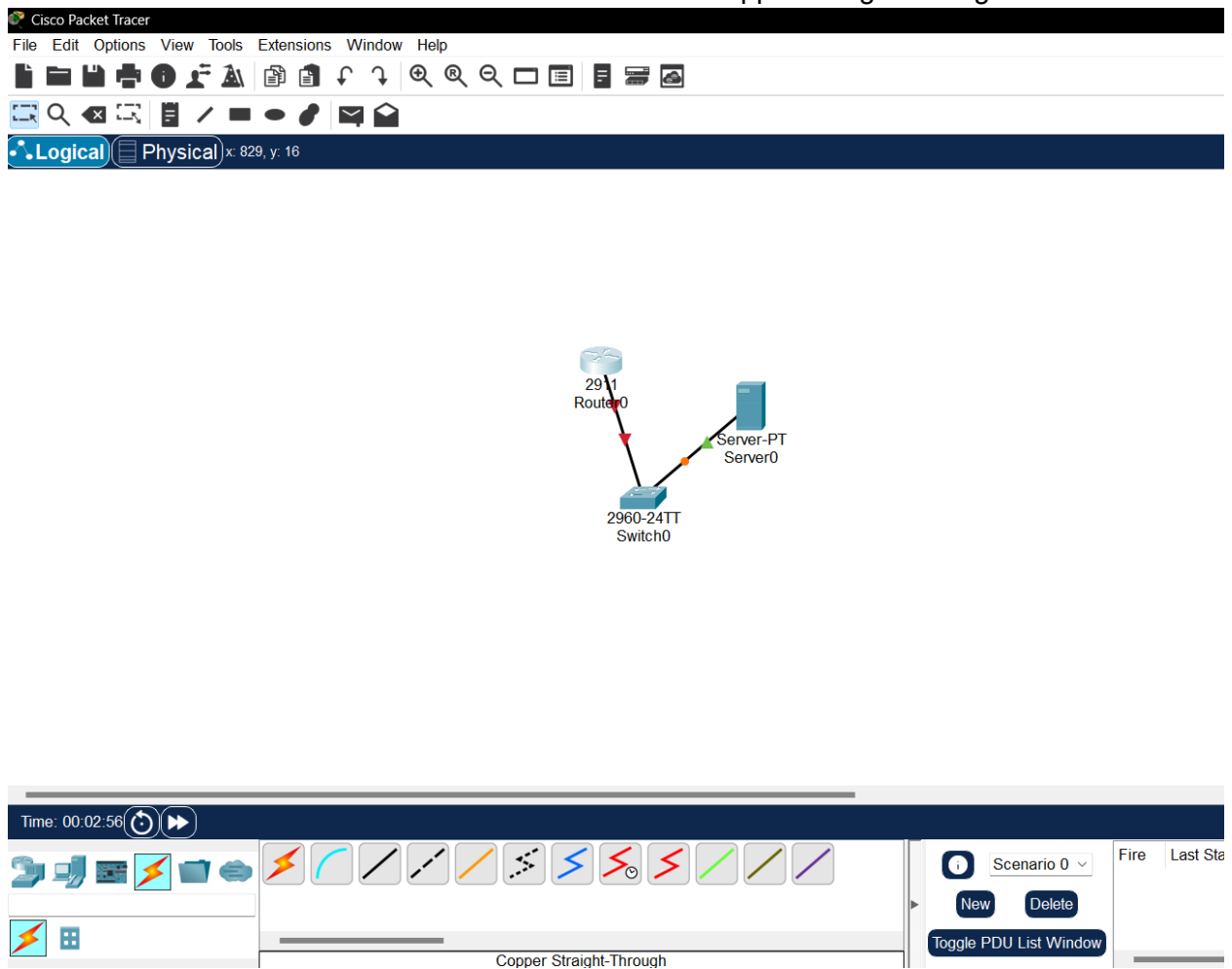
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5/13//2023

Update IOS(ManagementServer)

1. Connected a 2960 switch to server and 2911 router via copper straight-through cable



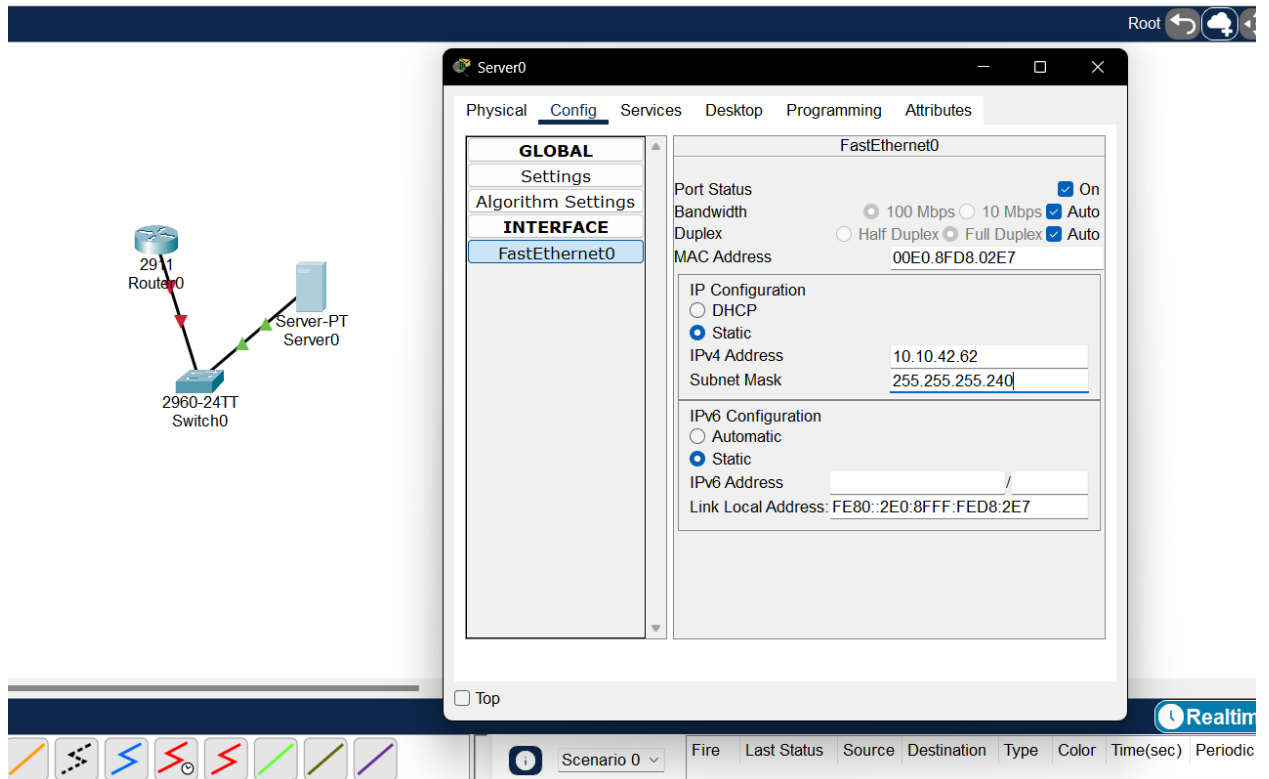
2. Configuring ip address 10.10.42.60 for the router port g0/0

The screenshot displays a network simulation interface. On the left, a topology diagram shows a router labeled '2911 Router0' connected to a switch labeled '2960-24TT Switch0', which is in turn connected to a server labeled 'Server-PT Server0'. The main window is titled 'Router0' and contains a terminal window with the following text:

```
export@cisco.com.  
  
Cisco CISCO2911/K9 (revision 1.0) with 491520K/32768K bytes of  
memory.  
Processor board ID FTX152400KS  
3 Gigabit Ethernet interfaces  
DRAM configuration is 64 bits wide with parity disabled.  
255K bytes of non-volatile configuration memory.  
249856K bytes of ATA System CompactFlash 0 (Read/Write)  
  
--- System Configuration Dialog ---  
  
Would you like to enter the initial configuration dialog? [yes/no]:  
  
Press RETURN to get started!  
  
Router>en  
Router#conf t  
Enter configuration commands, one per line. End with CNTL/Z.  
Router(config)#int g0/0  
Router(config-if)#ip address 10.10.42.82  
% Incomplete command.  
Router(config-if)#ip address 10.10.42.60 255.255.255.240  
Router(config-if)#
```

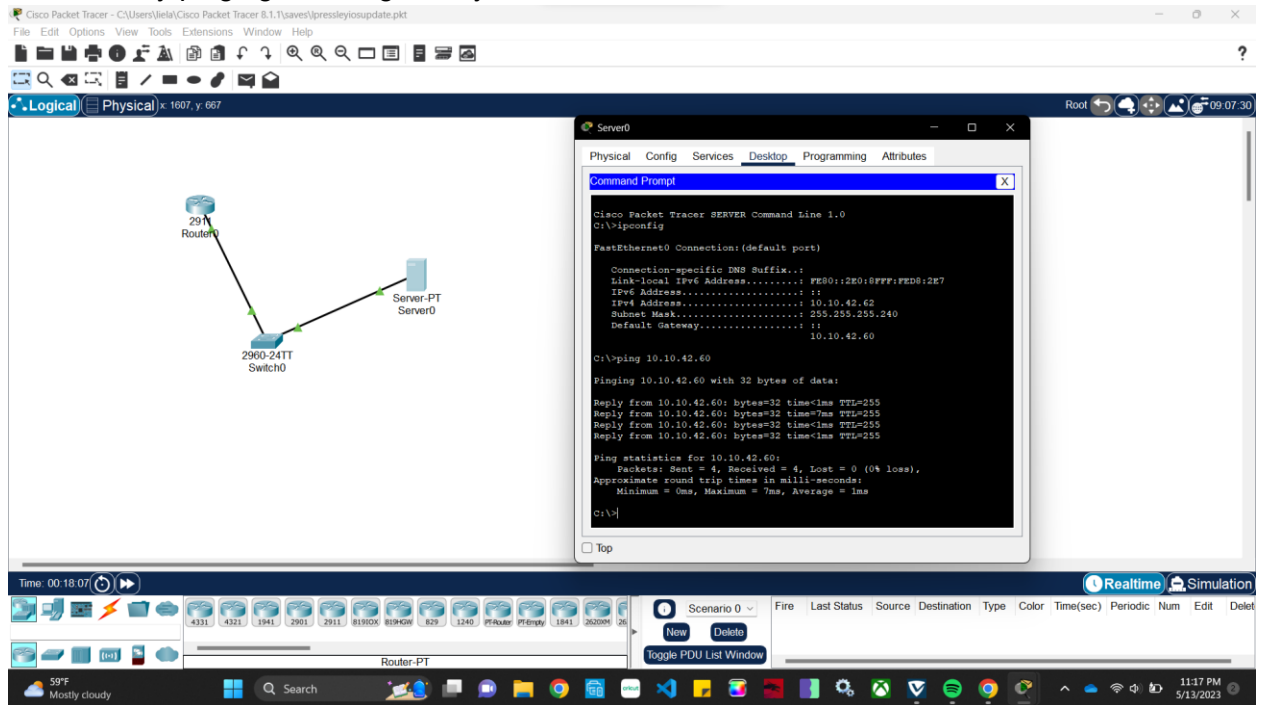
Below the terminal window, there are buttons for 'Copy' and 'Paste', and a 'Top' button. The bottom of the interface shows a toolbar with various icons and a status bar with the text 'Scenario 0' and a table with columns: 'Fire', 'Last Status', 'Source', 'Destination', 'Type', 'Color', 'Time(sec)', and 'Periodic'.

3. Configure the dns server and static ip and default gateway



The image shows a Cisco Packet Tracer network diagram and the configuration window for Server0. The network diagram on the left shows a 2911 Router connected to a 2960-24TT Switch, which is connected to a Server-PT (Server0). The configuration window for Server0 is open, showing the 'Config' tab. Under the 'INTERFACE' section, 'FastEthernet0' is selected. The 'IP Configuration' is set to 'Static' with an IPv4 Address of 10.10.42.62 and a Subnet Mask of 255.255.255.240. The 'IPv6 Configuration' is set to 'Static' with an IPv6 Address of FE80::2E0:8FFF:FED8:2E7 and a Link Local Address of FE80::2E0:8FFF:FED8:2E7.

4. Successfully ping the default gateway 10.10.42.60



The image shows the same Cisco Packet Tracer network diagram as above, but with the 'Command Prompt' window open on Server0. The Command Prompt shows the following commands and output:

```
Cisco Packet Tracer SERVER Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection (default port)
    Connection-specific DNS Suffix...: 
    Link-local IPv6 Address . . . . .: FE80::2E0:8FFF:FED8:2E7
    IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: 10.10.42.62
    Subnet Mask . . . . .: 255.255.255.240
    Default Gateway . . . . .: 10.10.42.60

C:\>ping 10.10.42.60

Pinging 10.10.42.60 with 32 bytes of data:

Reply from 10.10.42.60: bytes=32 time=1ms TTL=255
Reply from 10.10.42.60: bytes=32 time=1ms TTL=255
Reply from 10.10.42.60: bytes=32 time=1ms TTL=255
Reply from 10.10.42.60: bytes=32 time=1ms TTL=255

Ping statistics for 10.10.42.60:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 7ms, Average = 1ms

C:\>
```

5. Add an IEP address to server by creating default gateway vlan 1 address

Logical Physical x: 381, y: 304 Root

Switch0

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#ip default-gateway
% Incomplete command.
Switch(config)#ip default-gateway 10.10.42.60
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
wr
Building configuration...
[OK]
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int vlan 1
Switch(config-if)#ip address 10.10.42.61 255.255.255.240
Switch(config-if)#no shut

Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state
to up
exit
Switch(config)#exit
Switch#
```

Ctrl+F6 to exit CLI focus

Copy Paste

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4331 4321 1941 2901 2911 819IOX 819HGW 829 1240 PT-Router PT-Empty 1841

819HG-4G-IOX

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6. Successfully added the vlan and default gateway and successfully ping the server from the switch- fully established connection

The screenshot shows a network simulator window titled "Switch0" with tabs for "Physical", "Config", "CLI", and "Attributes". The "CLI" tab is active, displaying the "IOS Command Line Interface".

```
!
interface FastEthernet0/22
!
interface FastEthernet0/23
!
interface FastEthernet0/24
!
interface GigabitEthernet0/1
!
interface GigabitEthernet0/2
!
interface Vlan1
 ip address 10.10.42.61 255.255.255.240
!
 ip default-gateway 10.10.42.60
!
!
!
!
!
line con 0
!
line vty 0 4
 login
line vty 5 15
 login
!
!
!
!
end

Switch#ping 10.10.42.62

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.42.62, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms

Switch#ping 10.10.42.62

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.42.62, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms

Switch#
```

At the bottom of the CLI window, there is a status bar with the text "Ctrl+F6 to exit CLI focus" and two buttons: "Copy" and "Paste".

On the right side of the simulator, there is a system tray area showing the "alttime" logo, a "Simulati" label, and several icons representing different components: "router", "PI-Empty", "1841", and "2620XM".

7. Show what is currently in storage tftp file for 10.10.42.62

The screenshot shows a Cisco IOS Command Line Interface (CLI) window titled "Switch0". The window has tabs for "Physical", "Config", "CLI", and "Attributes", with "CLI" currently selected. The CLI shows the following sequence of commands and outputs:

```
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#ping 10.10.42.62

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.42.62, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

Switch#copy tftp flash
Address or name of remote host []? 10.10.42.62
Source filename []? c2960-lanbasek9-mz.150-2.SE4.bin
Destination filename [c2960-lanbasek9-mz.150-2.SE4.bin]?

Accessing tftp://10.10.42.62/c2960-lanbasek9-mz.150-2.SE4.bin...
Loading c2960-lanbasek9-mz.150-2.SE4.bin from 10.10.42.62:
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
[OK - 4670455 bytes]

4670455 bytes copied in 0.049 secs (7663014 bytes/sec)
Switch#dir
Directory of flash:/

  1  -rw-   4670455          <no date>  2960-lanbasek9-mz.
150-2.SE4.bin
  4  -rw-   4670455          <no date>  c2960-lanbasek9-mz.
150-2.SE4.bin
  3  -rw-    1128          <no date>  config.text

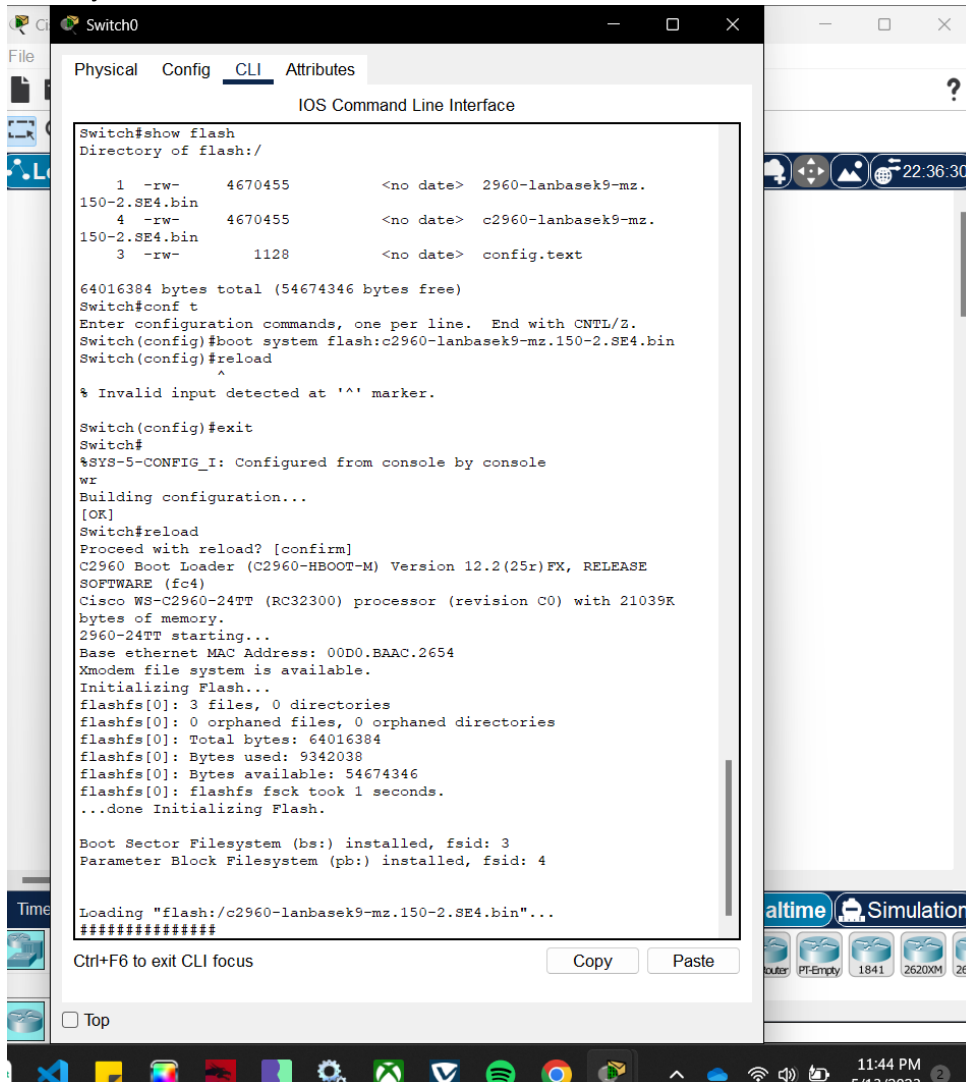
64016384 bytes total (54674346 bytes free)
Switch#show flash
Directory of flash:/

  1  -rw-   4670455          <no date>  2960-lanbasek9-mz.
150-2.SE4.bin
  4  -rw-   4670455          <no date>  c2960-lanbasek9-mz.
150-2.SE4.bin
  3  -rw-    1128          <no date>  config.text

64016384 bytes total (54674346 bytes free)
Switch#
```

At the bottom of the CLI window, there is a status bar that says "Ctrl+F6 to exit CLI focus" and two buttons: "Copy" and "Paste".

8. Boot system and reload the switch



Switch0

Physical Config **CLI** Attributes

IOS Command Line Interface

```
Switch#show flash
Directory of flash:/

 1 -rw-      4670455      <no date>  2960-lanbasek9-mz.
150-2.SE4.bin
 4 -rw-      4670455      <no date>  c2960-lanbasek9-mz.
150-2.SE4.bin
 3 -rw-         1128      <no date>  config.text

64016384 bytes total (54674346 bytes free)
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#boot system flash:c2960-lanbasek9-mz.150-2.SE4.bin
Switch(config)#reload
^
% Invalid input detected at '^' marker.

Switch(config)#exit
Switch#
$SYS-5-CONFIG_I: Configured from console by console
vr
Building configuration...
[OK]
Switch#reload
Proceed with reload? [confirm]
C2960 Boot Loader (C2960-HBOOT-M) Version 12.2(25r)FX, RELEASE
SOFTWARE (fc4)
Cisco WS-C2960-24TT (RC32300) processor (revision C0) with 21039K
bytes of memory.
2960-24TT starting...
Base ethernet MAC Address: 00D0.BAAC.2654
Xmodem file system is available.
Initializing Flash...
flashfs[0]: 3 files, 0 directories
flashfs[0]: 0 orphaned files, 0 orphaned directories
flashfs[0]: Total bytes: 64016384
flashfs[0]: Bytes used: 9342038
flashfs[0]: Bytes available: 54674346
flashfs[0]: flashfs fsck took 1 seconds.
...done Initializing Flash.

Boot Sector Filesystem (bs:) installed, fsid: 3
Parameter Block Filesystem (pb:) installed, fsid: 4

Loading "flash:/c2960-lanbasek9-mz.150-2.SE4.bin"...
#####
```

Ctrl+F6 to exit CLI focus

Copy Paste

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alttime Simulation

router PT-Empty 1841 2620XM 26

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9. You can see that we successfully updated the file

The screenshot shows the CLI of a Cisco Switch0. The interface displays various system details including hardware specifications, MAC address, and serial numbers. A table lists the switch ports, showing 1 port of type 26 with model WS-C2960-24TT-L, running software version 15.0(2)SE4. The interface also shows the software image C2960-LANBASEK9-M. At the bottom, there is a message indicating the configuration was set from the console by console.

Physical Config **CLI** Attributes

IOS Command Line Interface

compliance with U.S. and local country laws. By using this product you agree to comply with applicable laws and regulations. If you are unable to comply with U.S. and local laws, return this product immediately. A summary of U.S. laws governing Cisco cryptographic products may be found at: <http://www.cisco.com/wwl/export/crypto/tool/stqrg.html> If you require further assistance please contact us by sending email to export@cisco.com.
cisco WS-C2960-24TT-L (PowerPC405) processor (revision B0) with 65536K bytes of memory.
Processor board ID FOC1010X104
Last reset from power-on
1 Virtual Ethernet interface
24 FastEthernet interfaces
2 Gigabit Ethernet interfaces
The password-recovery mechanism is enabled.
64K bytes of flash-simulated non-volatile configuration memory.
Base ethernet MAC Address : 00:D0:BA:AC:26:54
Motherboard assembly number : 73-10390-03
Power supply part number : 341-0097-02
Motherboard serial number : FOC10093R12
Power supply serial number : AZS1007032H
Model revision number : B0
Motherboard revision number : B0
Model number : WS-C2960-24TT-L
System serial number : FOC1010X104
Top Assembly Part Number : 800-27221-02
Top Assembly Revision Number : A0
Version ID : V02
CLEI Code Number : COM3L00BRA
Hardware Board Revision Number : 0x01

Switch	Ports	Model	SW Version	SW Image
*	1 26	WS-C2960-24TT-L	15.0(2)SE4	C2960-LANBASEK9-M

Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 15.0(2)SE4, RELEASE SOFTWARE (fc1)
Technical Support: <http://www.cisco.com/techsupport>
Copyright (c) 1986-2013 by Cisco Systems, Inc.
Compiled Wed 26-Jun-13 02:49 by mnnguyen

Press RETURN to get started!

%SYS-5-CONFIG_I: Configured from console by console

Ctrl+F6 to exit CLI focus

Copy Paste

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